



Significance of predictors: revisiting stock return predictions using explainable AI

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Abstract

In this paper, we re-examine 166 previously identified asset pricing characteristics and their ability to successfully predict stock returns. We use Explainable Artificial Intelligence to rank these return predictors based on their importance in various asset pricing model settings. Our findings suggest that ensemble and deep learning-based models have an advantage in providing generalized predictions across different return measures. Using SHapley Additive exPlanations, we also find that momentum and trading-based features possess higher predictive power in estimating asset returns. The long-short portfolio analysis reveals that key return predictors exhibit substantial economic significance, reflected in the large differences in out-of-sample R^2 . These findings remain robust across various models and persist even after controlling for characteristics-based predictors.

Keywords Asset pricing · Machine learning · Explainable AI · Model predictability · FinTech · SHAP

1 Introduction

The use of machine learning (ML) in finance, particularly in asset pricing, has surged rapidly in recent years. This growth is primarily due to the ability of these approaches to handle large datasets with complex non-linear interactions. Compared to traditional econometric approaches, ML models are better equipped to capture multi-way predictor interactions and identify non-linear effects. Several empirical studies show that ML, particularly deep learning (DL) models, can significantly enhance asset return predictability and portfolio performance (Gu et al., 2020, 2021; Chen et al., 2024; Leippold et al., 2022; Uddin & Yu, 2020).

The black-box nature of ML models presents significant challenges in interpreting their performance and understanding their predictions. In finance, both practitioners and policymakers are often interested in understanding the contribution of various predictors to

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a model's overall performance. Although some ML models, such as decision trees, offer built-in predictor importance, they are far from ideal and often provide limited information on model explainability. In DL models, a large number of non-linear interactions makes it difficult to estimate which predictors contribute the most to return predictability and under what conditions. Therefore, creating explainable DL models, along with achieving performance improvements, is essential.

Highly sophisticated ML models are often complex and opaque, leading to increased interest in model explainability within the Artificial Intelligence (AI) research community. Researchers are focused on understanding the underlying mechanisms of these algorithms and ensuring that their predictions can be effectively applied in practice. Several frameworks have been developed to address these challenges and improve model explainability. One of the most widely used frameworks for explaining model predictability is SHapley Additive exPlanations (SHAP), developed by Lundberg and Lee (2017) based on Shapley values (Shapley & Shubik, 1954). SHAP offers an intuitive framework for introducing model explainability in this rapidly growing field of empirical asset pricing via ML. It enables us to address fundamental questions, such as how to interpret complex black-box models and identify the characteristics or features that play the most significant role in explaining returns. SHAP assigns each feature an importance score based on the marginal contribution it makes in predicting the outcome variable. Feature importance is expressed in the units of the dependent variable, allowing for clear and efficient interpretation. In addition to local explainability, SHAP offers generalized global explainability of features, providing a comprehensive view of how they interact and contribute to the final predictions of any ML model.

In this paper, we apply the SHAP framework to various ML models to identify key characteristics that significantly contribute to predicting cross-sectional asset returns. We construct ML models using 166 previously identified asset pricing predictors along with 12 macroeconomic factors. These ML models vary in complexity, ranging from simple linear regression (LR) to non-linear ensemble models, such as Random Forests (RF), Extra Tree Regressor (XTR), and Extreme Gradient Boosting Tree (XGB) to more complex Deep Neural Networks (DNN) with multiple hidden layers. We also examine the performance variations of these models under different characteristic controls, including the market factor, Fama and French (1993)'s three-factor model, and Carhart (1997)'s four-factor model.

The primary motivation for using SHAP as the main measure of explainability is its capacity to capture non-linear interactions and interdependencies among predictors— an essential characteristic of financial data. This capability affords several advantages over other ad hoc methods for calculating feature importance among asset pricing predictors (Chen et al., 2024; Gu et al., 2020). First, the feature importance evaluation method proposed by Chen et al. (2024) relies on the specific structure of a no-arbitrage constraint-based generative adversarial network. In contrast, SHAP is model-agnostic and can be applied to any ML model, enhancing its versatility. Second, unlike the iterative retraining required by the feature importance calculation method proposed by Gu et al. (2020), in which the model must be refitted for each feature, SHAP generally requires only a single model fit. As a result, SHAP offers considerable computational efficiency. Finally, gradient-based methods for assessing feature importance in neural network models, such as Sadhwani et

al. (2021); Horel and Giesecke (2020), assume that features are independently distributed. This assumption is often invalid for complex financial datasets, where variables frequently display substantial interdependencies and correlations, potentially undermining the reliability of such methods in detecting true feature interactions. By contrast, SHAP does not rely on such assumptions and allocates feature importance consistently across various feature combinations, enabling it to capture both global and local interactions more accurately. As a result, SHAP offers a more robust explainability framework, well-suited to explain the intricate dependencies within financial data.

To ensure robustness, we employ two additional methods to assess explainability. The first is an iterative feature importance approach based on Gu et al. (2020). In this method, the model is iteratively fitted on the data with one feature value set to zero (henceforth, referred to as leave-one-out method or LOOM). The aggregated change in model performance, measured by the out-of-sample R^2 score (R_{OOS}^2), reflects the importance of the feature. The second method involves constructing long-short portfolios based on individual feature values. We then analyze the variation in R_{OOS}^2 across these portfolios. A significant difference in R_{OOS}^2 between long and short portfolios for a given feature underscores its relevance to the predictive performance of the ML model.

Our work revealed several interesting findings. First, we observe that boosting algorithms provide better out-of-sample performance. The ability of these models to learn sequentially from their residuals makes them a better choice when tuned adequately using grid search. Second, we find that many predictors are interoperable when predicting various excess returns. Momentum and trading-based features consistently rank high in feature importance across various methods. Third, we observed that the feature importance derived from SHAP exhibits slight differences compared to that obtained through the LOOM method. Features such as beta (market beta) and roaq (quarterly return on assets) that were ranked important by the SHAP approach, deemed unimportant according to LOOM. However, the portfolio analysis shows that the most important predictors identified by SHAP also exhibit significant long-short portfolio R_{OOS}^2 highlighting the economic significance of these predictors. Finally, we found that the results are consistent when controlling for market, size, value, and momentum. After controlling for the market, some market-related factors, such as beta, became less significant; however, we did not observe any notable reduction in the importance of momentum-related factors, even after introducing the Carhart (1997) momentum factor as a control.

The contributions of this paper are multi-fold. First, we conducted a comprehensive analysis of an extensive feature set and provided a list of open-source features that exhibit high predictive power when coupled with specific ML algorithms. The paper offers explanations for the selection of models and features, along with a holistic comparison of performance across various algorithms. These findings can be used to develop custom-built ML models with fewer, yet important, predictors without sacrificing performance. Second, we highlight the use of the SHAP's model explainability framework to generate explainability at both the global model and local prediction level. We find that SHAP is less computationally intensive than LOOM. The latter approach requires fitting a new model for each feature, which becomes highly challenging as the feature space grows, particularly with the addition of interaction terms with macroeconomic factors. Third, through interaction term and dependency analysis, we demonstrate how feature interactions affect model performance. These

findings provide researchers and decision-makers with valuable insights into the interdependence of characteristics and macro predictors, and their influence on future asset returns.

This work also makes a significant contribution to the literature on the application of ML in asset pricing, especially in the area of explainable AI. Earlier studies have employed firm-specific characteristics and macroeconomic predictors as inputs for various ML models, including Auto-encoders (Gu et al., 2021; Uddin & Yu, 2020) and Multi-layer Perceptrons (MLP) (Gu et al., 2020; Leippold et al., 2022; D’Amato et al., 2021), to forecast future excess returns. Additionally, some researchers utilize unstructured data, such as news (Atkins et al., 2018) and social media (Houlihan & Creamer, 2021), to gauge market sentiment and predict stock returns (Naeem et al., 2021). Beyond equity markets, ML-based models have been applied to predict pricing dynamics in other asset classes, including options (Goudenege et al., 2020; Eckstein et al., 2021), mortgage-backed securities (Deku et al., 2021), cryptocurrencies (Jana et al., 2021; Akyildirim et al., 2021), and real estate (Jiang et al., 2022; Drobetz & Otto, 2021). Most of these studies primarily focus on achieving higher predictive accuracy.

However, recently there has been a growing shift in focus from improving prediction accuracy to enhancing model explainability (Zhang et al., 2023; Del Vitto et al., 2023; Jabeur et al., 2024). Explainable AI techniques, such as AutoML and SHAP along with gradient-based methods as seen in Sadhwani et al. (2021); Horel and Giesecke (2020) have been successfully applied in fields like real-estate, healthcare (Leung et al., 2022), engineering (Nyawa et al., 2023), and transportation (Veran et al., 2020) to analyze algorithmic fairness and model trustworthiness. Despite the increasing application of ML in finance, the use of explainable AI, particularly in asset pricing, remains limited. This paper addresses this gap by providing insights into feature and model selection processes in empirical asset pricing using ML, thus offering enhanced explainability.

The remainder of the paper is organized as follows: Section 2 provides a detailed description of our models and data, along with the evaluation framework used for model assessment. Section 3 presents our main empirical results, accompanied by a comprehensive discussion of the findings. Section 4 reports the results of the robustness tests. Finally, Section 5 offers concluding remarks and summarizes the key insights of the paper.

2 Data and methodology

2.1 Data

In this paper, we use the open-source cross-sectional asset pricing data from Chen and Zimmermann (2022). This dataset comprises monthly data from December 1925 to June 2022, incorporating 210 predictor variables that have been extensively documented in the literature as significant predictors of cross-sectional asset returns.¹

We perform some essential data preprocessing before feeding the data into the ML models. Several firms and predictors exhibit a significant number of missing values, particularly in the earlier years. To address this issue, we limit the sample period to January 2000 through July 2019 for our main experiment. Second, among all existing firms and indicators,

¹ The complete set of predictor data and detailed descriptions are available on Andrew Y. Chen’s website at <https://sites.google.com/site/chenandrewy/>.

we retain only those with more than 50% non-missing observations. Third, we eliminate individual rows that contain 70% or more missing values. Finally, for the remaining missing data, we apply a missing value imputation technique. We impute each feature using forward-filling when a firm has a non-missing observation in the previous time period; otherwise, we impute using the class median for that feature. After preprocessing, the final dataset consists of 10,075 firms and 166 predictors.

In addition to firm-specific predictors, we also incorporate global macroeconomic factors as inputs to our ML models. Following Gu et al. (2020), we include 12 macroeconomic characteristics identified by Welch and Goyal (2008) in our feature set. These macro characteristics are dividend-price ratio (dp), dividend yield (dy), earnings-price ratio (ep), dividend payout ratio (de), equity risk premium volatility (rvol), and many others.² However, unlike Gu et al. (2020), rather than using all raw macroeconomic features, we follow Ludvigson and Ng (2009) by calculating latent factors from macroeconomic characteristics. To calculate these latent factors, we perform principal component analysis (PCA) on the macro characteristics and select the top three components (henceforth referred to as PC1, PC2, and PC3), as their cumulative explained variance is 99.9%. We then use these latent factors to compute interaction terms with the firm-specific predictors, which serve as input features. This approach provides two key benefits. First, it helps to avoid multicollinearity in the macroeconomic data. Second, it allows us to retain a more compact feature space, as the use of interaction terms can increase the feature space exponentially.

Ultimately, the total number of covariates, $z_{i,t}$, is $166 \times (3 + 1) = 664$; comprising 166 firm-specific characteristics and three latent macroeconomic factors. The interaction variables are named according to the latent components that interact with the firm characteristics to create the final feature set. For instance, when PC2 interacts with the firm feature MaxRet (Maximum return over a month), the resulting feature is designated as MaxRet_2.

For all ML models, we use the next month's return as the dependent variable. In line with existing asset pricing literature, we employ both excess return and abnormal return as measures of return. Excess return is calculated by subtracting the Treasury bill rate from the log return. Abnormal return over the market (w/R_m) is obtained by estimating market model parameters for each firm using a 60-month rolling window, with a minimum requirement of 30 months.

To enable a comprehensive comparison across models and their identified predictors, we conduct multiple robustness tests. These include extending the data timeline (January 1990 to July 2019) and performing a moving window analysis with alternative time intervals. Additionally, we experiment with randomly split train-test datasets to evaluate the stability of our results across different time periods and training configurations. Finally, we assess each model's performance on abnormal returns derived from the Fama-French three-factor model (FF3), which incorporates size and value factors alongside the market factor, and the Carhart four-factor model (FF4), which adds momentum factor to FF3.³ Abnormal returns from FF3 and FF4 are calculated similarly to market abnormal returns, employing a 60-month rolling window with a minimum requirement of 30 months.

²The details of these macroeconomic characteristics are available on Amit Goyal's website at <https://sites.google.com/view/agoyal145>.

³Data for the market, FF3, and FF4 models are obtained from Kenneth French's website at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

2.2 Models

We employ a battery of ML techniques to develop predictive models for cross-sectional asset pricing. Following the empirical design established by Gu et al. (2020), we construct these ML models. The future excess return of an asset can be defined as an additive error model:

$$r_{i,t+1} = E_t(r_{i,t+1}) + \epsilon_{i,t+1}, \quad (1)$$

where,

$$E_t(r_{i,t+1}) = g^*(z_{i,t}). \quad (2)$$

Here, the function $g^*(\cdot)$ indicates the conditional expected return that uses the predictors $z_{i,t}$ of firm i at time t to estimate the excess return $r_{i,t+1}$ of next month $t + 1$. For estimating $g^*(\cdot)$, we consider Linear Regression (LR), Random Forest (RF) (Ho, 1995), Extra Tree Regressor (XTR) (Akbari et al., 2021), Extreme Gradient Boosting (XGB) (Chen & Guestrin, 2016), and Deep Neural Network (DNN) with 1 to 5 hidden layers.⁴

We select these specific models because they represent a broad spectrum of ML techniques, ranging from simple ensemble models, such as RF, to more complex DNN. This diversity in methodologies allows us to leverage their unique strengths and capabilities, thereby providing a robust foundation for predictive modeling in cross-sectional asset pricing. The selected models offer a well-balanced approach between explainability (e.g., RF) and complexity (e.g., DNN). By integrating models of varying complexity, we effectively address the trade-off between accuracy and explainability, fostering a more comprehensive understanding of asset pricing dynamics. In addition, these models are widely utilized as benchmarks in contemporary asset pricing literature (Chen et al., 2024; Uddin et al., 2023), with studies such as Gu et al. (2020), Gu et al. (2021), and Leippold et al. (2022) demonstrating their superiority over linear models in predicting cross-sectional asset returns. In the following sections, we provide detailed descriptions and implementation specifics for these models.

Linear Regression (LR) The simplest and most intuitive model we employ is LR via ordinary least squares (OLS). Owing to its simplicity and ease of interpretability, OLS is widely used across various research domains. We perform a multivariate LR using OLS, where all predictors serve as independent variables. This can be expressed by the following equation:

$$g^*(z_{i,t}; \theta) = Z'_{i,t} \theta, \quad (3)$$

where $g^*(\cdot)$ is estimated as a linear function of the predictors $z_{i,t}$ and parameters θ .

Random Forest (RF) RF is a robust ML model known for its efficiency and ease of implementation (Breiman, 2001). It operates based on the concept of bootstrap aggregation (bagging), an ensemble method in which n decision trees are trained on bootstrap samples from the training set, and their predictions are aggregated to generate the final output. The key

⁴The Python implementation of the source code is available at <https://github.com/BhaskarGoswami1998/Significance-of-Predictors-Revisiting-Stock-Return-Predictions-Using-Explainable-AI.git>

hyperparameters considered for RF include the number of estimators ($n_estimators$), representing the individual decision trees, and the maximum depth of each tree (max_depth), which helps mitigate overfitting. The bagging approach reduces variance and enhances the model's ability to generalize effectively on test data. For our study, we employ the RF algorithm from the scikit-learn library.

Extra Tree Regressor (XTR) Introduced by Geurts et al. (2006), XTR is an ensemble learning method designed to create de-correlated trees from random subsets of the data (without replacement). The results from each tree are aggregated to produce the final prediction. The XTR is similar to the RF but with a key difference, it selects the split points completely at random, rather than choosing the best possible split from the random subset of features (e.g., using the Gini index or entropy). XTR is generally faster than RF due to the random selection of splitting values rather than calculating them at every point. For our study, we employ the algorithm from the scikit-learn library.

Extreme Gradient Boosting Machine (XGB) XGB is a widely used gradient-boosting algorithm that builds on tree-based ML models (Chen & Guestrin, 2016). It employs a boosting technique, wherein weak learners (shallow decision trees) are grown depth-wise and iteratively refined to minimize training errors. By an ensemble of multiple regression trees, XGB generates a final prediction as the cumulative sum of the outputs from each individual tree. While highly effective, XGB is prone to overfitting, especially when the model complexity is high, necessitating careful tuning of hyperparameters such as learning rate, tree depth, and regularization parameters to achieve optimal performance. Its flexibility and scalability make it a popular choice for large-scale datasets. For our study, we use the algorithm as defined by the xgboost library.

Multi-layer Perceptron (MLP) MLP is a fully connected feed-forward neural network that comprises an input layer, output layer, and multiple hidden layers (Psaltis et al., 1988). Based on the number of hidden layers, MLP may be considered either a shallow or DNN. In this paper, we examined five different MLP settings-NN1 to NN5, with one (50 nodes), two (50 and 100 nodes), three (50, 100, and 150 nodes), four (150, 100, 50 and 25 nodes) and five (150, 100, 50, 25 and 12 nodes) hidden layers, respectively. For all models, the input layer's dimensionality corresponds to the size of the feature space, denoted as P . The output layer consists of a single node, responsible for predicting the dependent variable $r_{i,t+1}$.

For each hidden layer l , the output is calculated recursively by aggregating the output of each neuron k of the previous layer $l - 1$ as $x_k^{(l)} = \sigma(x^{(l-1)'} \theta_k^{(l-1)})$. The final output function of the neural network is defined as:

$$g(z; \theta) = x^{(L-1)'} \theta^{(L-1)} \quad (4)$$

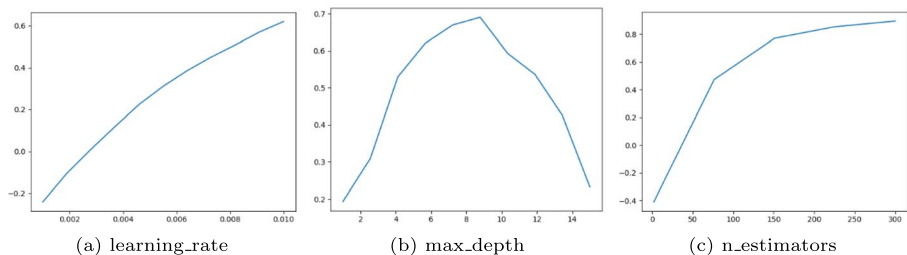
where L is the total number of hidden layers. σ denotes the activation function. For each hidden layer, we use the rectified linear unit (ReLU) as defined in Equation 5 as the nonlinear activation function. We use a single neuron with a linear activation function for the final output layer.

$$ReLU\phi(x) = \begin{cases} 0 & x \leq 0 \\ x & x > 0 \end{cases} \quad (5)$$

Table 1 Hyperparameters

Models	Hyperparameter	Values	Reported Parameters
LR	—	—	—
XTR	n_estimators	5–200	100
	max_depth	2–15	5
	max_features	5–150	50
RF	n_estimators	5–100	50
	max_features	5–130	50
	max_depth	1–9	5
XGB	n_estimators	5–200	100
	max_depth	2–15	5
	learning_rate	1e-3, 5e-3, 1e-2, 5e-2	1e-2
NN1	learning_rate	1e-2, 1e-3, 1e-4	1e-2
	batch_size	64, 120, 150, 200	64
NN2	learning_rate	1e-2, 1e-3, 1e-4	1e-2
	batch_size	64, 120, 150, 200	64
NN3	learning_rate	1e-2, 1e-3, 1e-4	1e-2
	batch_size	64, 120, 150, 200	64
NN4	learning_rate	1e-5, 5e-5, 1e-4	1e-5
	batch_size	120, 150, 200, 1000	120
NN5	learning_rate	1e-5, 5e-5, 1e-4	1e-5
	batch_size	120, 150, 200, 1000	120

This table presents the hyperparameters utilized in all the ML models. The range of values examined is displayed in column three, while column four indicates the parameter values of the final model that yield the results reported in this paper

**Fig. 1** Effect of hyper-parameters (x-axis) on R^2_{OOS} score (y-axis) in XGB

The hyperparameters for the aforementioned models are detailed in Table 1. In column three, we present the range of values examined, while column four indicates the final parameter values used to estimate the results reported in this paper. Figures 1, 2, and 3 depict the impact of hyperparameters on XGB, RF, and XTR, respectively. The horizontal axis represents the hyperparameters, and the vertical axis shows the out-of-sample R^2_{OOS} score. For the XGB and XTR models, we observed an inverted U-shaped pattern for the max_depth parameter, indicating that increasing model depth did not necessarily improve model performance. Additionally, we noted that for several other hyperparameters, such as n_estimator

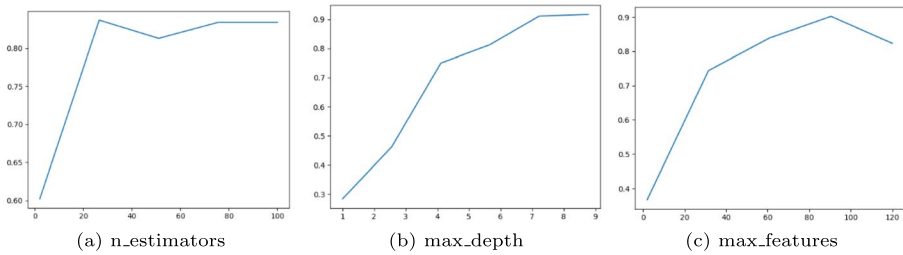


Fig. 2 Effect of hyper-parameters (x-axis) on R^2_{OOS} score (y-axis) in RF

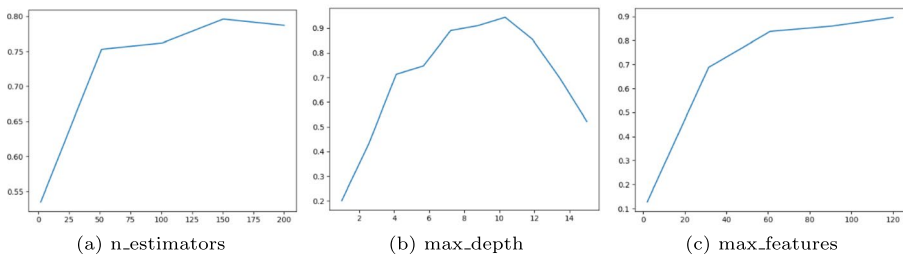


Fig. 3 Effect of hyper-parameters (x-axis) on R^2_{OOS} score (y-axis) in XTR

and `max_features`, performance improvement began to plateau after a certain point. This observation suggests diminishing marginal returns in model performance and implies that further improvements may be subsumed by the increased complexity associated with incorporating more features or estimators into the model.

2.3 SHapley additive exPlanations (SHAP)

Once the ML models are built, we employ the explainable AI framework, SHAP, to understand the importance of individual predictors in both local predictions and global model settings. Introduced by Lundberg and Lee (2017), SHAP is based on Shapley values (Shapley & Shubik, 1954) and is used to assign each feature an importance score based on its marginal contribution to the predictions. To compute the Shapley value for a feature i in the context of an ML model $g^*(\cdot)$, we consider all possible combinations of features. The Shapley value for the feature i , denoted as ϕ_i , is given by:

$$\phi_i = \sum_{S \subseteq Z \setminus \{i\}} \frac{|S|!(|Z| - |S| - 1)!}{|Z|!} [g^*(S \cup \{i\}) - g^*(S)] \quad (6)$$

here, $|Z|$ is the total number of features in the model, S is the subset of features that does not include feature i , $|S|$ is the number of features in the subset S , $g^*(S)$ is the model's prediction using the features in subset S and $g^*(S \cup \{i\})$ is the model's prediction using the features in subset S along with feature i . This essentially calculates the marginal contribution of feature

i in the prediction performance of model $g^*(\cdot)$, and the fraction term $\frac{|S|!(|Z|-|S|-1)!}{|Z|!}$ represents the weight assigned to the marginal contribution of feature i when added to the coalition S . The weights in this calculation depend on two factors: the total number of possible combinations of features that have already been added and the number of combinations in which the remaining features can participate in the coalition.

The importance of each feature in SHAP is expressed in the units of the dependent variable, ensuring that the assessment of feature importance reflects their contributions to predictions when considered alongside other features. In addition to providing local explainability, SHAP offers a generalized global perspective on feature importance by averaging across samples after accounting for absolute differences. This approach delivers a comprehensive understanding of how features interact and contribute to the final prediction.

While Shapley values provide valuable insights into feature contributions, the computation of these values is computationally intensive, particularly for high-dimensional data. This complexity arises from the necessity to evaluate all possible feature combinations, which makes the approach less feasible for very large datasets. Nevertheless, Shapley values explain the extent to which each feature contributes to predicting a specific instance and how this contribution deviates from the average prediction across the dataset. By quantifying feature importance in this manner, SHAP enhances explainability and fosters a deeper understanding of the model's decision-making process.

2.4 Evaluation framework

To evaluate the performance of ML models, we divided the data into train and test subsamples on a temporal scale. Records between 2000 and 2017 were put in the training subsample to train the ML models. Data from 2018 and 2019 were included in the test sample. The purpose of dividing the data on a temporal scale is to avoid the problem of information leakage, a common issue in time-series prediction. This study does not incorporate a validation set, which is typically employed for hyperparameter tuning, as the primary objective is not to achieve the most accurate predictions. Instead, we focus on demonstrating the application of SHAP and a diverse array of commonly utilized ML models to enhance model explainability. This approach allows us to emphasize the interpretative capabilities of these models, thereby contributing to the broader understanding of their functionalities within the context of predictive analytics.

To evaluate the predictive performance of each model, we employ three widely recognized performance metrics: Out-of-Sample R^2 score (R_{OOS}^2), Mean Absolute Error (MAE), and Mean Squared Error (MSE). The formulas for these evaluation metrics are as follows:

$$MAE = \frac{1}{TN_{\in\Omega}} \sum_{(i,t) \in \Omega} |r_{i,t+1} - \hat{r}_{i,t+1}| \quad (7)$$

$$MSE = \frac{1}{TN_{\in\Omega}} \sum_{(i,t) \in \Omega} (r_{i,t+1} - \hat{r}_{i,t+1})^2 \quad (8)$$

$$R_{OOS}^2 = 1 - \frac{\sum_{(i,t) \in \Omega} (r_{i,t+1} - \hat{r}_{i,t+1})^2}{\sum_{(i,t) \in \Omega} (r_{i,t+1})^2} \quad (9)$$

here, T indicates the total number of time periods, N indicates the number of cross-sections, and Ω indicates the test sub-sample. A higher value of R_{OOS}^2 and a lower value of MSE and MAE indicate better performance.

3 Results and discussion

In our study, we analyze the models and predictors across three dimensions. First, in Subsection 3.1, we evaluate each model's accuracy on the training and testing subsamples in predicting excess returns. In Subsection 3.2, we report the significance of individual predictors across models using SHAP and compare the result with LOOM (Gu et al., 2020). Lastly, in Subsection 3.4, we discuss the importance of predictors based on the performance of long-short portfolios.

3.1 Model selection based on predictive accuracy

We start by analyzing the accuracy of each model in predicting cross-sectional asset returns with and without market controls. In Table 2, Panel A reports the model performances when trained to estimate cross-sectional excess returns over the risk-free rate, whereas Panel B

Table 2 Predictive accuracy of ML models

Models	Train			Test		
	MSE	MAE	$R_{OOS}^2(\%)$	MSE	MAE	$R_{OOS}^2(\%)$
<i>Panel A: Excess Return (No Control)</i>						
LR	227.124	9.893	1.303	237.421	9.988	−0.258
XTR	232.028	10.166	1.019	249.882	10.630	0.513
RF	224.003	10.001	4.443	250.705	10.676	0.185
XGB	222.278	9.962	5.179	249.416	10.681	0.698
NN1	127.617	7.108	0.604	404.556	10.133	0.114
NN2	102.841	6.762	0.681	376.697	9.687	0.175
NN3	154.478	8.199	0.521	422.462	10.709	0.075
NN4	229.204	10.115	2.224	250.105	10.657	0.424
NN5	228.540	10.107	2.507	250.001	10.658	0.465
Average	194.235	9.146	2.053	299.027	10.424	0.266
<i>Panel B: Abnormal Return from the Market Model (w/R_m)</i>						
LR	205.436	9.512	1.794	217.889	9.731	0.278
XTR	207.254	9.533	0.925	216.786	9.702	0.783
RF	204.329	9.482	2.323	216.752	9.709	0.798
XGB	204.448	9.492	2.266	217.129	9.731	0.626
NN1	156.831	6.493	0.468	374.534	9.182	0.107
NN2	98.020	6.576	0.667	362.532	9.124	0.135
NN3	111.143	6.827	0.623	376.818	9.101	0.101
NN4	205.99	9.505	1.532	217.235	9.707	0.576
NN5	205.738	9.502	1.652	217.344	9.708	0.526
Average	177.668	8.547	1.361	268.558	9.522	0.437

This table reports the MSE, MAE, and $R_{OOS}^2(\%)$ for both the training and testing sets. Panel A presents the results for excess returns without any control, while Panel B provides the prediction results for abnormal returns, controlling for the Market model

presents the models trained to estimate abnormal returns derived from market models. In the table, the first three columns present the MSE, MAE, and R^2_{OOS} values for the training data, while the last three columns display these metrics for predicting the test data. Additionally, we provide the average performance across all models in the final row of each panel.

In our empirical analysis, we observe that all models exhibit a significant drop in performance when transitioning from training to test data. For instance, there is an 18% drop for the XTR and 27% for RF in panel A. Panel B demonstrates a similar trend with the NN1, NN2, and NN3. This general decline in performance highlights the issue of overfitting commonly encountered in complex ML models. Previous studies have documented a similar phenomenon in the application of ML to financial data, particularly with ensemble models like XGB, which attempt to fit residuals at each iteration (Uddin et al., 2022). Hyperparameter tuning and regularization can be employed to mitigate this overfitting issue.

In the test subsample, ensemble models such as XTR, RF, and XGB demonstrate superior performance in predicting both excess and abnormal returns. LR exhibits poor R^2_{OOS} performance, likely due to its inability to capture the complex, non-linear properties of data within a high-dimensional feature space. Similar findings have been reported by Gu et al. (2020). The neural network-based models (NN1 to NN5) demonstrate consistent performance in predicting both excess returns and abnormal returns, with the deeper neural networks (NN4 and NN5) outperforming the shallower models (NN1, NN2, and NN3) (Kelly et al., 2024). Financial data, particularly cross-sectional asset pricing data, are complex and characterized by multiple interaction effects and latent patterns. These results suggest that more advanced and sophisticated methodologies are better suited for capturing these hidden and non-linear interactions.

The results presented in Table 2 indicate that neural network-based models are sub-optimal compared to ensemble methods. However, it is important to note that the neural network configurations utilized in this analysis are relatively shallow, with the deepest models consisting of only five layers and minimal regularization. For comparison, the well-known Transformer model used in natural language processing contains 12 hidden layers (Vaswani et al., 2017). More complex neural networks with additional hidden layers and neurons could potentially achieve superior performance. However, exploring all possible model configurations to identify the optimal architecture would exponentially increase the computational complexity of the task. Since identifying the best neural network is not the primary objective of this paper, we have opted to maintain a simple network structure.

3.2 Predictors importance using SHAP

As we documented in subsection 3.1, ML models are more effective in predicting cross-sectional asset returns. We now shift our focus to our main objective - identifying the significance of predictors. To accomplish this, we employ the SHAP framework (Lundberg & Lee, 2017), which measures the average marginal contribution of a feature to a model's outcome across all possible feature combinations. SHAP achieves this by iteratively creating data samples, holding a subset of features constant, and recording the difference between the observed prediction and the average prediction. The global feature importance is then calculated by considering all possible combinations of features and aggregating them into a single metric representing the mean of these differences.

We run global feature analysis using SHAP for the best-performing models as shown in Section 3.1. It is done for the setting while predicting the excess return from the market model. Figure 4 presents the top 20 most essential predictors in estimating future abnormal returns from the market model using SHAP while using the XTR model. For readability, the analysis for the other models and other dependent variables is presented in Appendix A. The figure illustrates the directional impact of a feature on the model output. Each dot represents a specific data instance, the horizontal axis shows the SHAP value of that feature (either positive or negative), and the color gradients indicate the original input feature values. The indicators are sorted in descending order based on their importance. The feature name followed by a number, e.g., 1, 2, 3, indicates the feature's interaction term with the principal component of the macro factors. For example, Beta_2 signifies that the market beta interacts with the second principal component of the macro factors.

According to the SHAP scores, the most influential factors on cross-sectional asset return predictability are DivOmit_1 (Dividend Omission interaction with Macro PC1) and Exch-

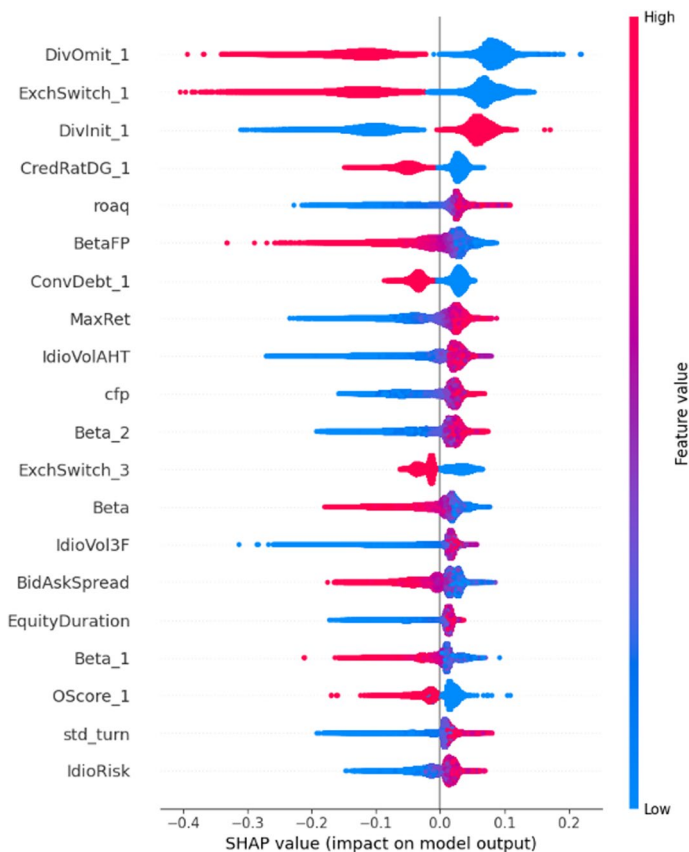


Fig. 4 This figure depicts the SHAP value of the top twenty predictors on the final test data. The predictors are sorted in descending order based on their importance (SHAP value). The predictor name followed by “_1, _2, or _3” denotes the interaction term of the feature with the first, second, or third principal component of the macroeconomic factors. The blue (red) demarcates the positive (negative) contribution of the feature on the final prediction

Switch_1 (interaction of Exchange Switch with Macro PC1). The directional contribution of these two factors is almost identical. These factors negatively correlate with model predictability, with lower values positively affecting estimated returns. However, the relationship is not strictly linear; rather, it is nonlinear and asymmetric. Negative values of these two predictors form a ball-shaped structure in the positive spectrum of SHAP values, whereas positive values are dispersed in a line. The third most influential predictor is DivInt_1 (Dividend Initiation), which shows a dense cluster of high values in the positive SHAP range, indicating a strong positive relationship with the model's outcome. Similar to the findings of Gu et al. (2020), our analysis also suggests that MaxRet (Recent Maximum Return) and roaq (Return on Assets Observed Quarterly) are among the most influential features in the model prediction process. Other important features include BetaFP (Frazzini-Pedersen Beta), ConvDebt (Convertible debt indicator), IdioVolAHT (Idiosyncratic volatility), cfp (Operating Cash flows to price), Beta (CAPM beta), BidAskSpread (Bid-ask Spread), EquityDuration (EquityDuration), Oscore (Ohlson's O-score introduced by Dichev (1998)) and idiorisk (Idiosyncratic risk).

Two noticeable patterns emerge from the SHAP values of these significant predictors. First, their impact on the model's outcome is nonlinear and asymmetric. The SHAP values range across both positive and negative spectra, with blue indicating positive effects and red indicating negative effects. Second, the positive impact tends to be more concentrated, as represented by the ball-shaped structures. We conduct SHAP analyses for other ensemble ML models, including RF and XGB, while also changing the outcome variable to other excess returns. The results are largely consistent, with minor variations in the ranking. Detailed results are provided in Appendix A.

In addition to global representations, SHAP allows for the analysis of model explanations at the local level for individual data points. Figure 5 demonstrates an example of local explanations for individual predictors within each ensemble model in the prediction of abnormal returns. The red (blue) areas represent a positive (negative) direction, while the thickness reflects the magnitude of the deviation from the average prediction value. This allows us to quantify the percentage change in the prediction of cross-sectional returns

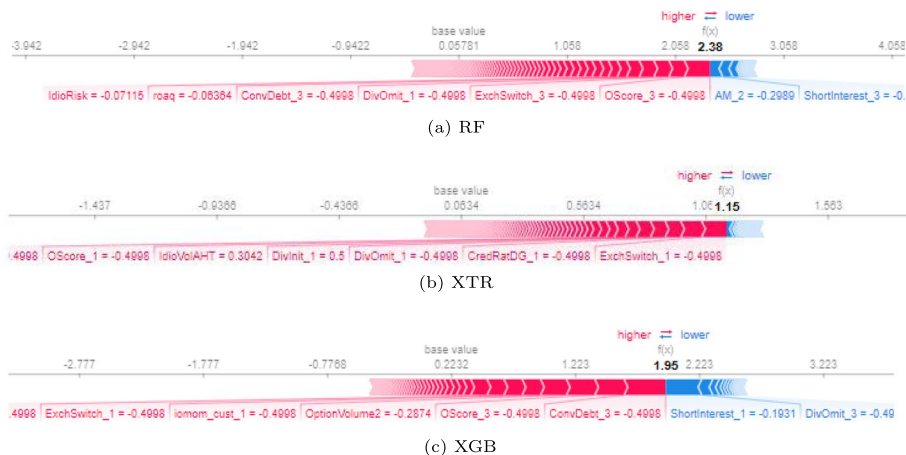


Fig. 5 This figure depicts a local prediction using SHAP in predicting abnormal returns from the market model

attributed to each feature. For example, in Fig. 5, the XGB model demonstrates that ConvDebt_3 (the interaction of Convertible Debt with macro PC3) and ShortInterest_1 (the interaction of Short Interest with macro PC1) contribute the most variance, shifting the base prediction from 0.22 to a final prediction of 1.95.

To understand the specific relationship of the predictors on the model outcome, we draw the dependency plots in Figs. 6 and 7. Figure 6 presents the top features in the model with no control, and Fig. 7 shows the model with control using the excess return from the market model. The horizontal axis of the dependency plot represents the indicator, the vertical axis indicates the SHAP value of the indicator, and the color gradient represents the interaction effect of the most influential feature on the original feature's performance.

The figures illustrate a non-linear relationship between the features and their interaction effects on the model outcomes. For example, subfig. 6a demonstrates a steep decline in SHAP values with increasing BetaFP (Frazzini and Pedersen (2014) Beta). This suggests that negative values for BetaFP contribute to positive SHAP values, while positive BetaFP values result in negative SHAP values. Additionally, we observe an interesting interaction effect with ChForecastAccrual (Change in Forecast Accrual) (Barth & Hutton, 2004). Higher values of ChForecastAccrual mitigate the influence of BetaFP on model performance, leading to a SHAP value that remains nearly constant and close to zero.

Similarly, in sub-fig. 6d, we observe that negative values of High52 (52-week high) positively influence the final prediction, with the effect plateauing at values close to one. Con-

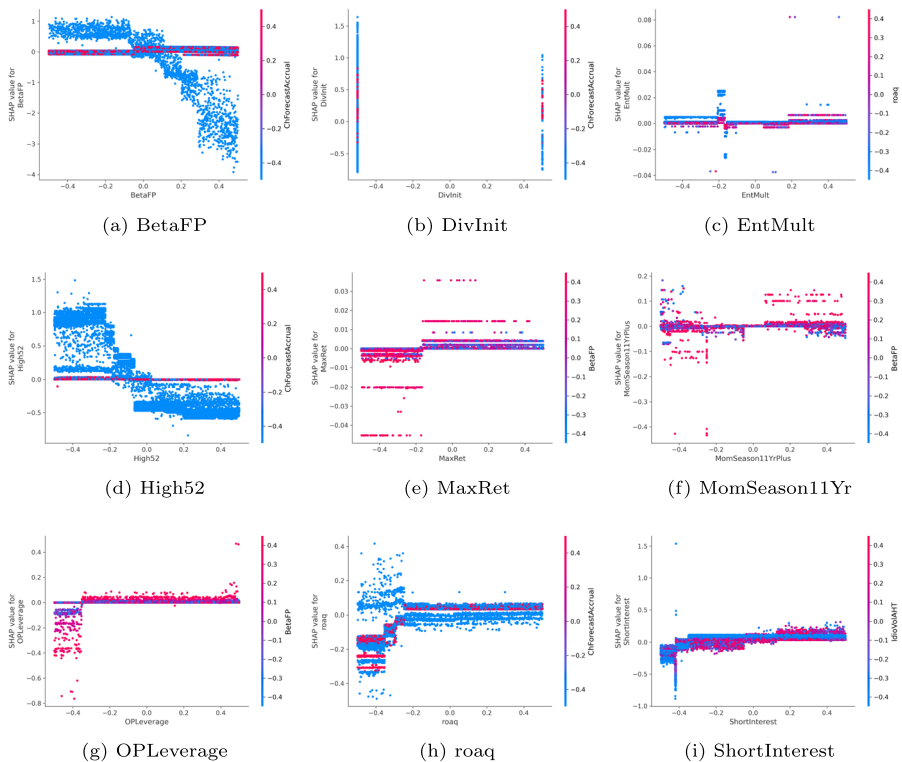


Fig. 6 This figure depicts the dependence plots of the top nine features for the XGB Model, showing their individual as well as interaction effects in predicting excess return with no controls

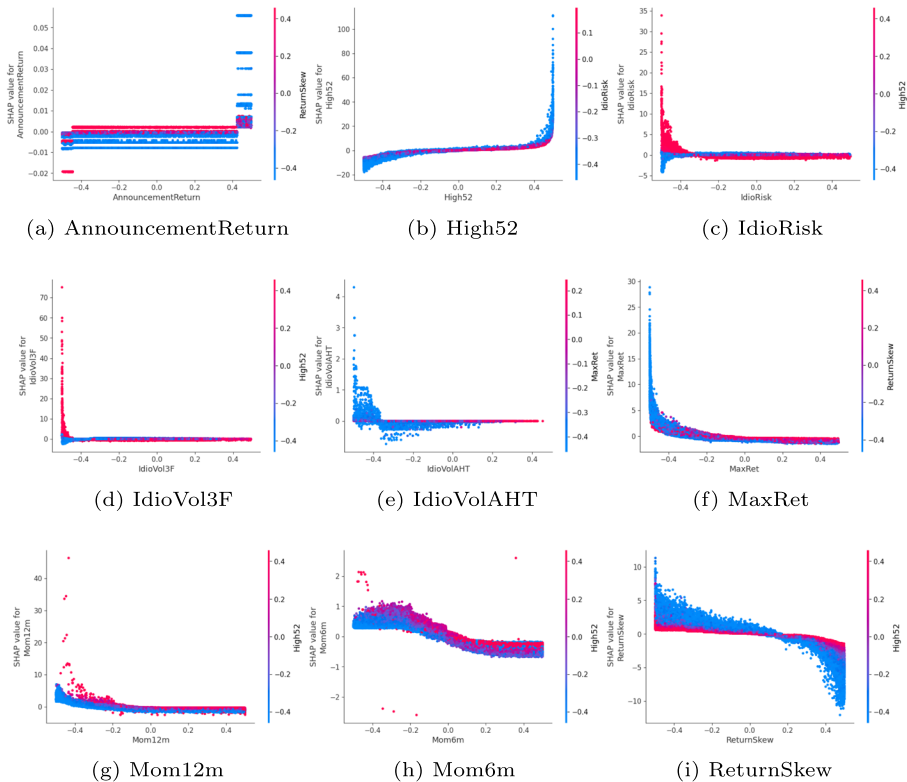


Fig. 7 This figure depicts the dependence plots of the top nine features for the XGB Model, showing their individual and interaction effects in predicting excess return after controlling for the Market factors

versely, for positive values of High52, negative SHAP values are recorded, resulting in final predictions that fall below the average prediction value. This effect flattens out at higher values. Additionally, we find that for higher values of change in forecast accrual, the effect is nullified, with SHAP values concentrated near zero.

For a comprehensive understanding, we present the overall results across models as a heat map of feature importance in Fig. 8. It illustrates all the predictors sorted based on their average contribution across different ML models. Figure 8a shows the heat map of models without control and Fig. 8b depicts the model with market control. We observe many common variables in the top part of each heat map, indicating that they are the most important predictors for the models in making accurate predictions. For example, the variables High 52 (52-week high), Mom12m (12 months Momentum), MaxRet, Mom6m (6 months momentum), volMkt (market volume) IdioRisk, IdioVolAHT, ReturnSkew (skewness) and Beta, appears in the top ten of every model for each dependent variable. This provides crucial insight into selecting important predictors in cross-sectional asset pricing.

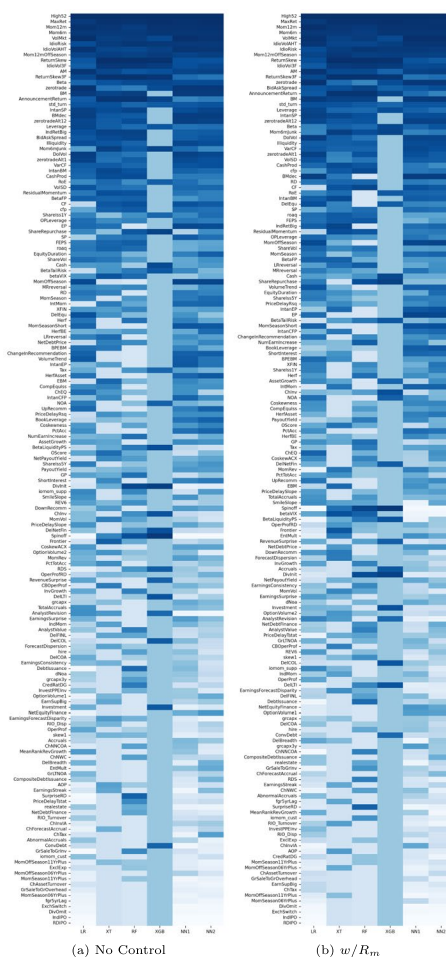
Fig. 8 These heatmaps illustrate the ranking of features based on their overall contribution to the ML models, calculated using SHAP (Lundberg & Lee, 2017). The color gradient represents the range from most important (dark blue) to least important (light blue)



3.3 SHAP Versus LOOM

In this section, we analyze feature importance based on the LOOM proposed by Gu et al. (2020) and compare it with SHAP. In their approach, Gu et al. (2020) iteratively refit the model with one feature set to zero at a time to assess each feature's impact on the model's out-of-sample performance. The aggregated change in the R^2_{OOS} score for each feature quantifies its importance, indicating the extent to which predictive accuracy declines when a given feature is excluded. We present the results of feature importance based on LOOM in Fig. 9. Figure 9a depicts the feature importance for models with no control, while Fig. 9b displays the model with market control. For each ML model, we refit the model after setting each feature under consideration to zero, then calculate the change in R^2_{OOS} . The values are standardized and ranked in descending order. The heat map represents the average feature importance across the various models analyzed.

Fig. 9 These heatmaps illustrate the ranking of features based on their overall contribution to the ML models, calculated using LOOM (Gu et al., 2020). The color gradient represents the range from most important (dark blue) to least important (light blue)



Our comparison of results from both methods reveals several interesting findings. Firstly, variables such as *mom1m* (Short-term Reversals), *mom12m* (Momentum), and *MaxRet* (Recent Maximum Return) exhibit high overall importance using the LOOM method. This aligns with findings reported by Gu et al. (2020), though their analysis involved a smaller feature set (94 features). Secondly, in our extensive feature set analysis using Gu et al. (2020)'s method, variables such as *Beta* (CAPM Beta) and *roaq* (Quarterly Return on Assets) rank within the top 50% of the importance list. However, these variables are not reported as among the most important predictors in Gu et al. (2020). This indicates that incorporating more predictors can influence the importance ranking of features, highlighting the significance of interaction effects in ML model performance. Lastly, by leveraging the open-source data from Chen and Zimmermann (2022), we identify several new features, including the industry return of large firms (*IndRetBig*) and the 52-week high, which exhibit higher importance when analyzed on a larger scale. Notably, this finding remains consistent even after controlling for market factors.

This provides a stable list of interoperable features across different dependent variables. It also shows that when we control for the market factors there is an aggregated drop of firm fundamentals like Beta. A summary list of the top ten features for each model and each dependent variable are presented in Appendix, Table 8. We further perform a comparative analysis of the feature ranking achieved from LOOM and SHAP. We look at Kendall's rank correlation coefficient (Kendall's τ) values, which is the measure of correspondence between two rankings. The τ is calculated as:

$$\tau = \frac{n_c - n_d}{\frac{1}{2}Z(Z - 1)}, \quad (10)$$

where, n_c refers to the number of concordant pairs, n_d refers to the number of discordant pairs, and Z refers to the total number of features including interaction terms. Table 3 reports Kendall's τ values across the two methods for the various ML models considered in our study. As shown in the table, the rankings of features across the models are notably similar, with Kendall's τ values ranging from 0.43 to 0.55, indicating a moderate to strong agreement between the methods. These minor differences are to be expected, as Kendall's τ coefficient is calculated based on exact pair matching. For instance, a feature ranked 10th in one method and 11th in another would be classified as discordant pairs in the calculation, despite their close proximity in ranking. However, in the context of feature importance in cross-sectional asset pricing, this difference is largely irrelevant. Nonetheless, the overall consistency in feature rankings suggests that both approaches provide converging insights into feature importance, underscoring their robustness in identifying influential predictors within the ML models.

In terms of model complexity, the LOOM method requires repeatedly fitting the model after setting the feature under consideration to zero in order to observe the change in R^2_{OOS} . Specifically, to calculate the change in R^2_{OOS} , it is required to fit exactly Z models, where Z represents the total number of covariates, including all predictors and their interaction terms. This iterative process results in significant model complexity and computational demand. Conversely, in most cases, SHAP requires the fitting of only a single model to com-

Table 3 Feature ranking comparison b/w LOOM and SHAP

		No Control		w/ R_m)		w/ $FF3$)		w/ $FF4$)	
		LOOM	SHAP	LOOM	SHAP	LOOM	SHAP	LOOM	SHAP
No control	LOOM	n/a	0.463	0.453	0.435	0.480	0.491	0.492	0.497
	SHAP		n/a	0.441	0.496	0.502	0.486	0.547	0.506
w/ R_m)	LOOM			n/a	0.451	0.464	0.471	0.494	0.474
	SHAP				n/a	0.463	0.479	0.467	0.502
w/ $FF3$	LOOM					n/a	0.471	0.492	0.501
	SHAP						n/a	0.453	0.547
w/ $FF4$	LOOM							n/a	0.472
	SHAP								n/a

This table reports Kendall's τ as a measure of similarity between feature rankings derived from the two methods: SHAP and LOOM. This metric ranges from -1 (complete disagreement) to +1 (complete agreement). The models examined include excess return with no controls (No Control), abnormal return from the market model (w/ R_m), abnormal return from the FF3 Model (w/ $FF3$), and abnormal return from the FF4 model (w/ $FF4$)

pute SHAP values. These values subsequently offer intuitive explanations of predictions and clarify feature interactions, making SHAP a more efficient choice in many scenarios.

3.4 Feature importance on portfolio estimation

In this experiment, we develop a long-short portfolio to analyze the top twenty predictors and their predictive importance. At each time period, we sort firms based on the value of the predictors, construct equally weighted quintile portfolios, and compute the out-of-sample R_{OOS}^2 for these portfolios. We update the portfolio constituents monthly based on the predictor's value in the previous months. For each indicator, the monthly average for the test period for each portfolio is reported in Table 4. Columns two to five report the average R_{OOS}^2 of five quintile portfolios, i.e., Low (L), 2, 3, 4, and High (H), respectively. Column six reports the difference between the high minus low (H-L) portfolio, and column seven reports the t-statistics of the high minus low portfolio.

The result is largely consistent with the findings of the SHAP. We find that the high-low portfolio of trading and momentum-related factors such as VolSD (volume variance), MaxRet (maximum return over a month), High 52 (52 weeks high), Mom12m (12 months momentum), and Mom6m_1 (six-month momentum) is statistically significant. A significantly positive (negative) high-minus-low value indicates that ML models are better

Table 4 Monthly portfolio-level out-of-sample predictive R_{OOS}^2

Predictor	Low (L)	2	3	4	High (H)	H-L	t-stat
VolSD	2.2620	1.3547	0.5928	0.1148	-0.2349	-2.4969	-3.7676*
MaxRet	1.7837	0.9286	0.2619	0.0071	-0.4524	-2.2361	-4.0205*
std_turn	2.0543	0.2378	-0.6034	-0.1335	-0.0215	-2.0758	-3.8726*
High52_1	2.0814	0.6014	0.2163	-0.2439	0.0548	-2.0266	-3.3655*
Beta	0.1254	0.0484	0.2747	0.7101	2.0058	1.8804	2.4949*
roaq	2.0299	0.4228	-0.1089	-0.0887	0.4549	-1.5749	-2.8121*
Mom12m	1.9970	0.4003	0.4922	0.3850	0.4980	-1.4990	-2.3811*
Mom6m_1	2.0033	0.6583	0.1759	0.4571	0.5277	-1.4756	-3.4129*
IdioVolAHT_2	1.1463	0.1243	-0.1418	-0.1338	0.0911	-1.0552	-1.4654
DolVol	0.1635	0.6422	0.7472	1.4683	1.1335	0.9700	1.9403*
fgr5yrLag_3	0.2835	0.9180	3.6757	-0.4977	1.0679	0.7844	2.8330*
Beta_2	0.7439	0.1961	0.2747	0.5625	1.3874	0.6436	0.8070
SP_1	1.6024	0.9614	0.6342	0.5974	0.9936	-0.6088	-1.2201
OptionVolume2	0.5764	1.1735	2.6657	0.8851	1.1612	0.5849	2.5140*
Mom12mOffSeason_1	1.5055	0.6367	0.4911	0.6664	0.9505	-0.5550	-1.4127
skew1	0.8271	1.4728	-4.3121	0.4161	0.3862	-0.4409	-1.6586
SmileSlope_1	1.2720	1.3595	0.4934	0.3011	0.9729	-0.2991	-1.1122
SmileSlope	1.2720	1.3595	0.4934	0.3011	0.9729	-0.2991	-1.1122
Leverage	0.9929	1.1504	0.9617	1.1119	1.1899	0.1970	0.3248
IndRetBig	0.7917	0.0328	1.1026	0.9386	0.9019	0.1102	0.2391
AnalystRevision_2	1.0720	1.4264	0.4626	0.4883	1.1652	0.0932	0.3457

This table reports the portfolio average predictive R_{OOS}^2 of the quintile portfolios for top-20 predictors. 'H-L' denotes the long-short portfolio that goes long in the portfolio of stocks with the highest feature value and short in the portfolio with the lowest feature value. t-stat is the t-statistics of 'H-L' portfolio. * indicate significance at 5% level. Here, the features are names as such, feature name followed by the PC number of the macro factors with which the term has interacted to form the final feature. If nothing is mentioned then it takes the original feature with no interaction

equipped to accurately predict the excess return for an asset with a high (low) value of that predictor. For example, a significant negative high-low value in VolSD indicates that the accuracy of the ML model varies depending on the volume variance, and the model accurately estimates the future excess return of an asset with a low volume variance. It is highly intuitive to anticipate that an asset pricing model would offer more accurate estimations for firms with a stable return structure compared to those with more volatile returns.

This long-short portfolio analysis suggests that the predictive accuracy of the ML model varies based on the feature values, and some features have an asymmetric impact on model performance. Significant differences in the R^2_{OOS} indicate that the model is highly sensitive to changes in feature values and demonstrate an asymmetric relationship with the model's predicted outcome. In contrast, features with insignificant R^2_{OOS} values indicate that the model is less sensitive to variability in those features. The findings also underscore the significance of interaction effects among asset pricing predictors on model performance. As illustrated, while several top predictors—such as VolSD, MaxRet, and std_turn—demonstrate substantial long-short performance independently, others, like Mom6m_1, SP_1, and IdioVolAHT_2, exhibit significance only when they interact with the Macro PC's.

4 Robustness test

4.1 Robustness tests with alternative return measures

In this section, we evaluate the robustness of our findings across multiple settings. First, we assess the robustness of the outcome variables by using two additional measures of abnormal returns. Specifically, we apply abnormal returns from the Fama and French (1993) three-factor model (w/FF3), which incorporates size and value factors alongside the market factor, and abnormal returns from the Carhart (1997) four-factor model (w/FF4), which adds momentum to the FF3 factors.

The result of the robustness test with these two abnormal returns is reported in Table 5. The overall ranking of the models remains consistent with the main analysis reported in Table 2. In both the FF3 and FF4 controls, nonlinear ensemble methods such as XTR, RF, and XGB outperform other models. Additionally, the DNN outperforms the shallow NN, further solidifying their superior predictive capacity in cross-sectional asset return. We also observe that there is an overall decline in performance after introducing these characteristics control. After introducing size and value as controls (Panel A), the R^2_{OOS} on test data dropped by 25, 22, and 19 basis points for XTR, XGB, and RF, respectively, compared to the market control model reported in Table 2 Panel B. Furthermore, after incorporating momentum (Panel B) to FF3, the R^2_{OOS} declined by an additional 33, 15, and 13 basis points for XTR, XGB, and RF, respectively. A similar pattern is observed in the DNN models. Compared to only market control, after controlling for FF3, the R^2_{OOS} declined by 26 and 14 basis points for NN4 and NN5, respectively. When we added momentum control, the R^2_{OOS} further declined by 15 and 17 basis points from the FF3 model. This general decline in performance is expected, as the four factors—market, size, value, and momentum—are widely regarded as the most dominant predictors of cross-sectional asset returns.

Table 5 Robustness test of predictive accuracy across ML models

Models	Train			Test		
	MSE	MAE	$R^2_{OOS}(\%)$	MSE	MAE	$R^2_{OOS}(\%)$
<i>Panel A: Abnormal Return from Fama and French Three-Factor Model (w/FF3)</i>						
LR	212.861	9.657	1.384	223.927	9.767	−0.046
XTR	214.551	9.677	0.601	222.651	9.701	0.524
RF	212.002	9.642	1.782	222.535	9.701	0.575
XGB	212.082	9.656	1.745	222.866	9.725	0.428
NN1	201.498	6.915	0.322	373.213	9.428	0.118
NN2	104.510	6.642	0.648	363.467	9.083	0.140
NN3	117.423	7.186	0.604	388.569	9.868	0.081
NN4	212.518	9.630	1.547	223.121	9.703	0.313
NN5	213.498	9.653	1.093	222.964	9.713	0.382
Average	188.994	8.740	1.081	273.701	9.622	0.279
<i>Panel B: Abnormal Return from Carhart Four-Factor Model (w/FF4)</i>						
LR	227.124	9.893	1.303	237.421	9.988	−0.258
XTR	228.87	9.913	0.544	236.354	9.925	0.192
RF	226.691	9.891	1.491	235.824	9.904	0.416
XGB	226.302	9.891	1.660	236.122	9.923	0.290
NN1	307.753	7.249	0.006	369.242	9.463	0.155
NN2	123.839	7.079	0.600	374.212	9.326	0.143
NN3	127.145	7.121	0.589	346.508	9.248	0.207
NN4	227.245	9.880	1.253	236.411	9.932	0.167
NN5	227.904	9.892	0.966	236.308	9.926	0.211
Average	213.653	9.979	0.935	278.711	9.737	0.169

This table reports the MSE, MAE, and $R^2_{OOS}(\%)$ of the ML models, utilizing alternative return measures—abnormal returns derived from the Fama and French Three-factor model (Panel A) and the Carhart Four-factor model (Panel B)

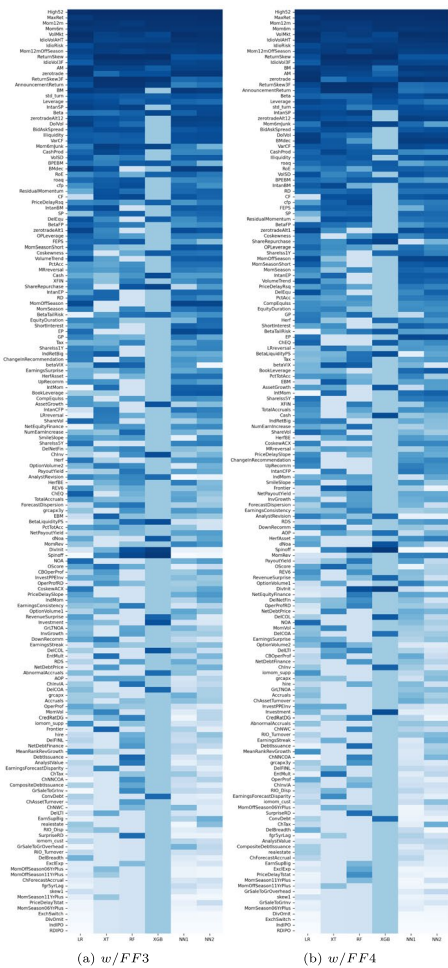
These factors have been extensively validated in the asset pricing literature for their ability to explain variations in returns across different portfolios. Consequently, when we control for their explanatory power and remove their contribution, the model's performance begins to decline.

Although the performance declines after controlling the outcome variable for FF3 and FF4, the importance of the predictors in cross-sectional asset returns remains consistent. We perform SHAP and LOOM feature importance analyses on the models using these two abnormal return measures. The heat map of the panel of all 166 main features sorted based on their average contribution to the final prediction is presented in Figs. 10 and 11. It shows consistent results when we compare them to the feature importance ranking from predicting excess return with no control and market control in Figs. 8 and 9.

4.2 Robustness test using random splitting and alternative time window

In our main analysis, we conduct a train-test split based on the temporal dimension, utilizing data from January 2000 to 2018 as the training set and data from 2018 to 2019 as the test set. As a result, our test set remains constant throughout the analysis, which raises concerns

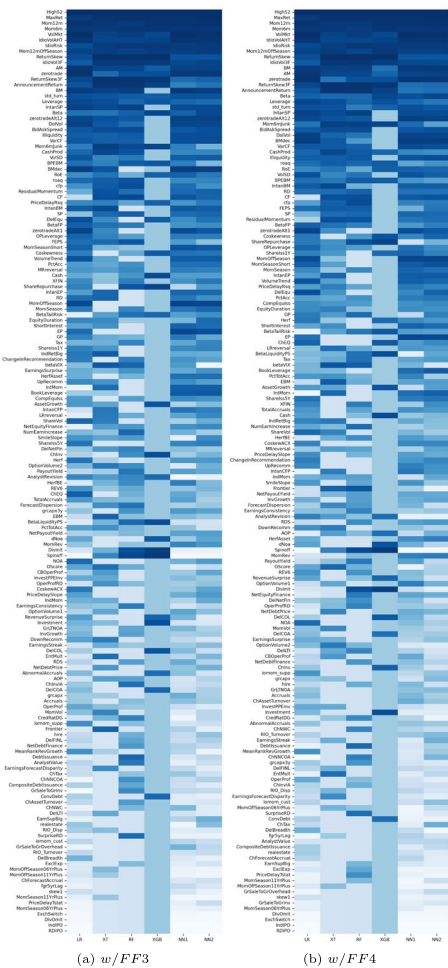
Fig. 10 These heatmaps illustrate the ranking of features based on their overall contribution to the ML models, calculated using SHAP. The color gradient represents the range from most important (dark blue) to least important (light blue)



about the temporal robustness of our framework. To demonstrate that our results are robust to different splitting techniques and alternative time windows, we perform two additional robustness tests. First, we compare model performance using a randomized train/test split keeping 90% in the train set and 10% in the testing subsample. Second, to further illustrate temporal robustness, we employ a multi-window framework with a five-year moving window.

Table 6 presents the results of the predictive performance of ML models on train-test sub-samples based on random splitting. The results are consistent with our initial analysis. In the majority of the cases, NN2 provides better and more generalizable results. We also notice that boosting and bagging-based models outperform DNN in both in-sample and out-of-sample settings. It is important to note that, for all models, the predictive R_{OOS}^2 values obtained from random splitting are significantly higher than those reported in our initial analysis in Table 2. This phenomenon is common in time-series predictions. Asset pricing

Fig. 11 These heatmaps illustrate the ranking of features based on their overall contribution to the ML models, calculated using LOOM (Gu et al., 2020). The color gradient represents the range from most important (dark blue) to least important (light blue)



data is multi-dimensional time-series data, where the temporal structure and relationships across variables play a critical role. A randomized split disregards these temporal dependencies and can introduce information leakage. When data from the future is included in the training set, the model gains access to information it would not have had in a real-world scenario, allowing it to estimate historical values based on future data points. This leads to artificially inflated performance metrics, as information from the training set inadvertently influences the test set, undermining the validity of the predictive evaluation.

In Table 7 we present the results from the multi-window analysis. We train our ML models—XGB, RF, XTR, NN1 to NN5—using the first four years of data and subsequently test the models in the following year. The objective is to evaluate the consistency of model performance over time and across various dependent variables, i.e., No Control, w/R_m , $w/FF3$, $w/FF4$. The results indicate consistent outcomes across different time windows when con-

Table 6 Predictive accuracy (R^2_{OOS}) for randomized split

	LR	XTR	RF	XGB	NN1	NN2	NN3	NN4	NN5
<i>Panel A: Excess Return (No Control)</i>									
Train	0.302	0.241	0.322	0.322	0.592	0.625	0.614	0.496	0.488
Test	0.317	0.252	0.321	0.329	0.562	0.547	0.550	0.490	0.457
<i>Panel B: Abnormal Return from Market Model (w/R_m)</i>									
Train	0.308	0.250	0.332	0.337	0.607	0.578	0.636	0.484	0.539
Test	0.324	0.263	0.332	0.344	0.562	0.537	0.555	0.472	0.516
<i>Panel C: Abnormal Return from Fama-French Three-Factor Model (w/FF3)</i>									
Train	0.288	0.235	0.314	0.320	0.568	0.600	0.521	0.482	0.564
Test	0.303	0.247	0.312	0.325	0.523	0.520	0.462	0.478	0.525
<i>Panel D: Abnormal Return from Carhart Four-Factor Model (w/FF4)</i>									
Train	0.277	0.227	0.305	0.307	0.549	0.584	0.565	0.487	0.520
Test	0.292	0.238	0.303	0.311	0.511	0.510	0.503	0.476	0.483

This table reports the R^2_{OOS} values of the ML models for both the training and testing sets, utilizing a random split. Panel A reports the excess return with no control, Panel B reports the abnormal return from the Market model, Panel C reports the abnormal return from the Fama and French Three-factor model, and Panel D reports the abnormal return from the Carhart Four-factor model

sidering combinations of ML models and dependent variables. However, notable variations in model performance are observed across time windows and model types. XGB consistently outperforms the other models, achieving its peak performance between 2000 and 2004. In contrast, the MLP-based models exhibit significant volatility, which increases with model complexity (a variance of 0.94 for NN1 and 1.12 for NN3). Other ensemble models, such as XTR and RF, demonstrate stable performance overall. The overly complex NN4 and NN5 models show negative R^2_{OOS} values, likely due to overfitting on a limited training period (4 years), resulting in poor performance on the test subset (1 year). This underperformance is observed consistently across different dependent variables, underscoring the need for larger datasets, precise hyperparameter tuning, and effective regularization when implementing deeper neural network models.

5 Conclusion

With the growing application of ML in finance, the demand for more explainability of these algorithms is also gaining rapid traction. Especially in the domain of asset pricing, it is imperative to understand the contribution of various predictors in stock return estimation and portfolio construction. This paper is an initial step in this regard. In this paper, we perform a comprehensive analysis of widely used asset pricing predictors through the optics of ML. The findings of this study reinforce our understanding of the importance of momentum and trading frictions in asset pricing. The results remain consistent even after controlling for multiple characteristics-based predictors and employing different approaches for robustness testing. The implications of these findings are manifold. They will assist researchers in narrowing down the search space for selecting optimal predictors and focusing on the most significant ones. Additionally, these findings can aid investors in developing profitable port-

Table 7 Predictive accuracy (R_{OOS}^2) across different time windows*Panel A: Excess Return (No Control)*

Years	XGB	RF	XT	NN1	NN2	NN3	NN4	NN5
1990-1994	0.282	0.220	0.274	0.150	0.193	0.012	−0.015	−0.015
1995-1999	0.346	0.335	0.272	0.528	0.580	0.591	0.000	0.000
2000-2004	0.447	0.428	0.405	−0.361	−0.512	−0.639	−0.005	−0.006
2005-2009	0.120	0.095	0.085	0.313	0.307	0.337	−0.032	−0.030
2010-2014	0.408	0.389	0.343	0.249	0.330	0.213	−0.006	−0.005
2015-2019	0.275	0.303	0.233	0.463	0.475	0.451	−0.005	−0.006
Average	0.292	0.270	0.260	0.207	0.187	0.224	−0.012	−0.011

Panel B: Abnormal Return from Market Model (w/ R_m)

1990-1994	0.294	0.233	0.288	−0.263	0.402	0.414	−0.003	−0.003
1995-1999	0.356	0.333	0.270	0.516	0.572	0.576	0.000	−0.001
2000-2004	0.490	0.459	0.413	−0.647	−0.401	−0.469	−0.003	−0.006
2005-2009	0.193	0.187	0.139	0.393	0.429	0.400	−0.004	−0.005
2010-2014	0.421	0.406	0.356	0.364	0.423	0.411	−0.001	−0.004
2015-2019	0.326	0.348	0.267	0.484	0.542	0.549	−0.001	−0.001
Average	0.298	0.306	0.265	0.058	0.344	0.379	−0.002	−0.003

Panel C: Abnormal Return from Fama-French Three-Factor Model (w/FF3)

1990-1994	0.275	0.230	0.275	0.211	0.241	0.390	0.000	−0.001
1995-1999	0.332	0.316	0.249	0.503	0.543	0.546	0.000	0.000
2000-2004	0.475	0.449	0.404	−0.175	−0.221	−0.379	0.000	−0.001
2005-2009	0.181	0.177	0.130	0.328	0.380	0.365	−0.006	−0.006
2010-2014	0.396	0.384	0.332	0.355	0.386	0.379	0.000	0.000
2015-2019	0.316	0.335	0.256	0.453	0.538	0.493	0.000	0.000
Average	0.309	0.308	0.274	0.232	0.287	0.299	−0.001	−0.001

Panel D: Abnormal Return from Carhart Four-Factor Model (w/FF4)

1990-1994	0.274	0.213	0.269	−0.464	0.017	−0.186	−0.002	−0.001
1995-1999	0.318	0.304	0.239	0.472	0.471	0.457	0.000	0.000
2000-2004	0.466	0.438	0.392	−0.356	−0.560	−0.666	−0.004	−0.003
2005-2009	0.159	0.158	0.119	0.312	0.315	0.302	−0.003	−0.002
2010-2014	0.392	0.379	0.331	0.360	0.307	0.124	0.000	−0.001
2015-2019	0.310	0.324	0.252	0.428	0.444	0.389	0.000	0.000
Average	0.287	0.264	0.260	0.097	0.172	0.086	−0.002	−0.001

This table reports the R_{OOS}^2 scores for the ML models across alternative time windows, utilizing different return measures - excess return without control (Panel A), abnormal return with market control (Panel B), abnormal return with Fama-French Three-factor control (Panel C), and abnormal return with Fama-Carhart Four-factor control (Panel D)

folios based on ML models. Lastly, this research offers an approach to incorporate transparency into overly complex ML algorithms and their integration into empirical asset pricing.

Appendix A: Result supplement

See Figs. 12, 13, 14, 15, 16, 17 and 18 and Table 8.

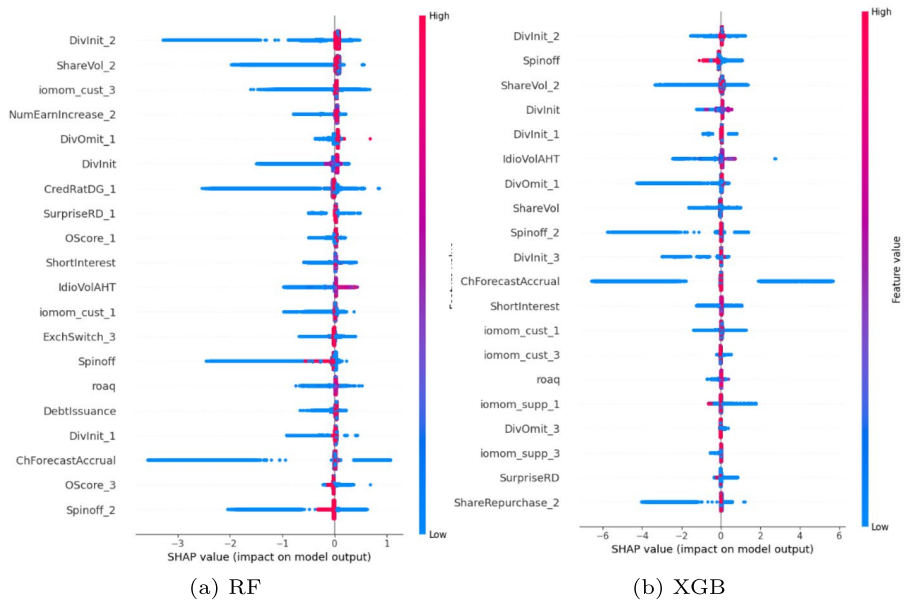


Fig. 12 SHAP values of the top twenty predictors in predicting excess return with no control using (a) RF and (b) XGB

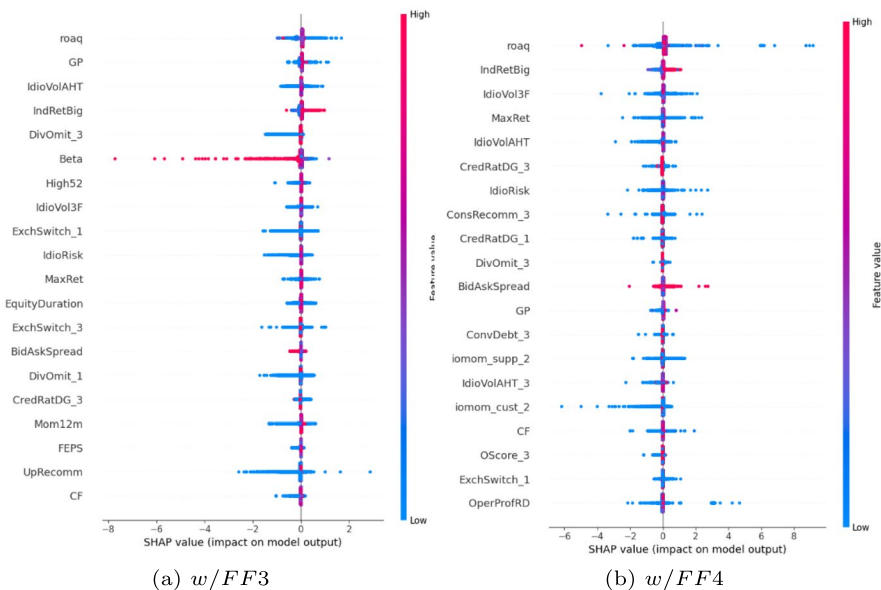


Fig. 13 SHAP values of the top twenty predictors using the RF model after controlling for (a) $FF3$ and (b) $FF4$ factors

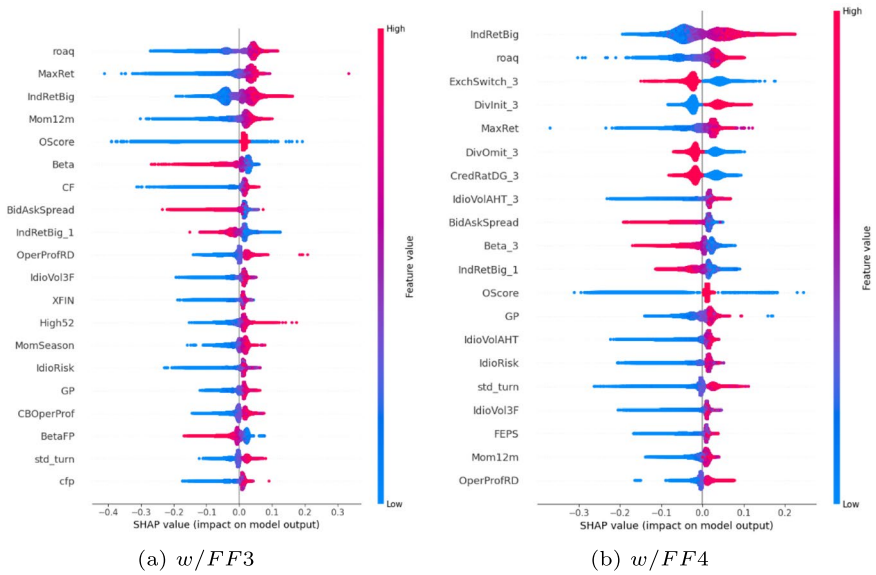


Fig. 14 SHAP values of the top twenty predictors using the XTR model after controlling for (a) $FF3$ and (b) $FF4$ factors

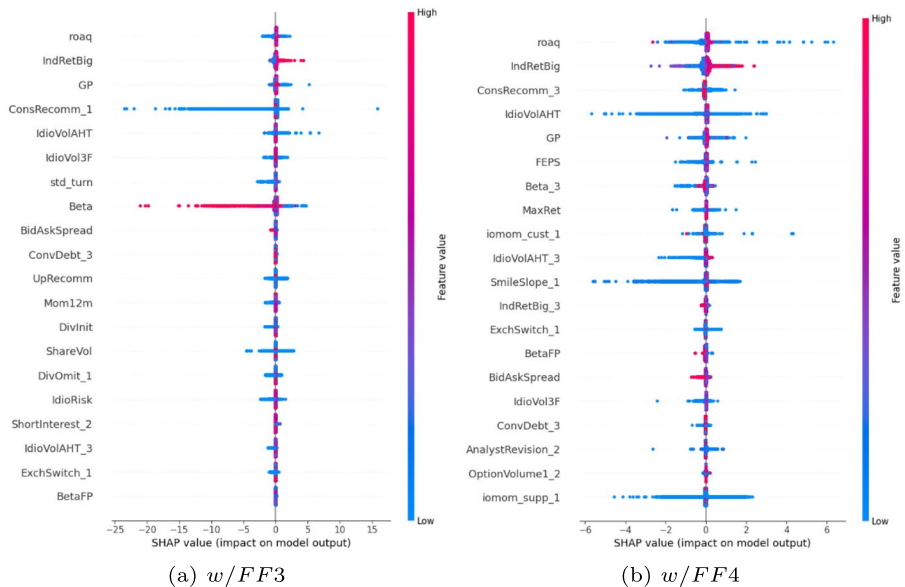


Fig. 15 SHAP values of the top twenty predictors using the XGB model after controlling for (a) $FF3$ and (b) $FF4$ factors

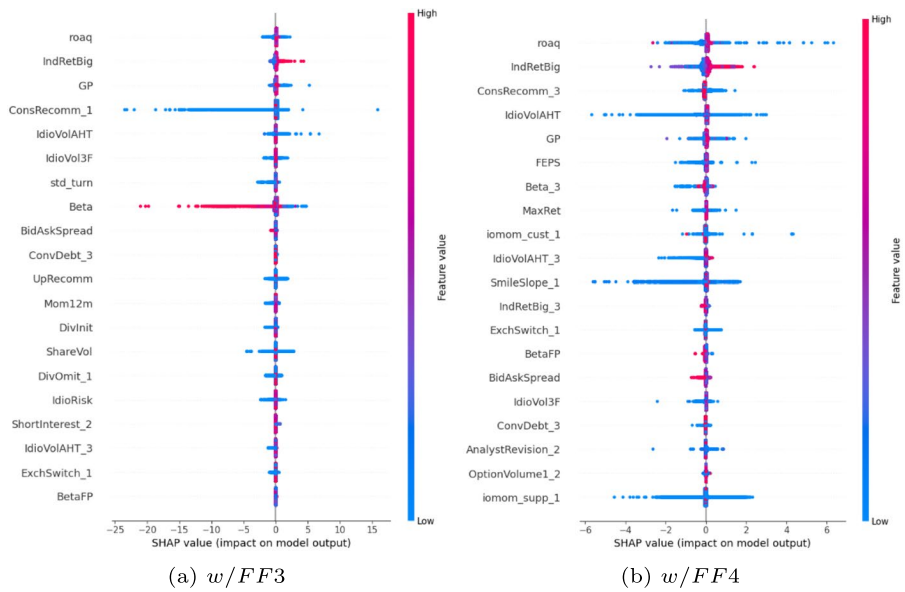


Fig. 16 SHAP values of the top twenty predictors using the XGB model after controlling for (a) $FF3$ and (b) $FF4$ factors

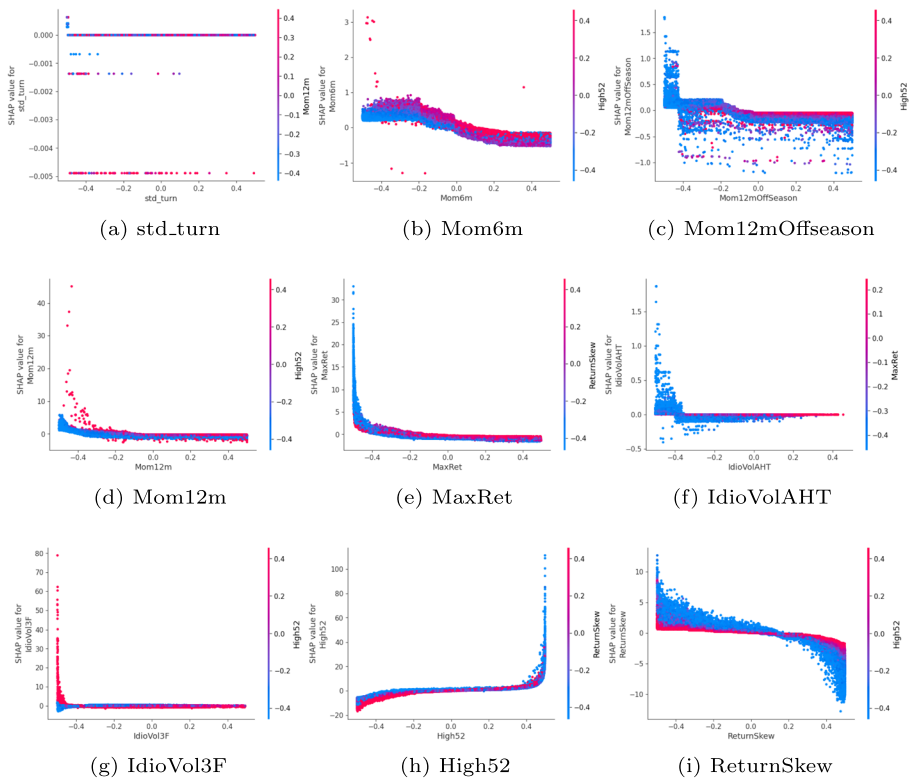


Fig. 17 This figure depicts the dependence plots of the top nine features for the XGB Model, showing their individual and interaction effects in predicting excess return after controlling for ($w/FF3$) factors

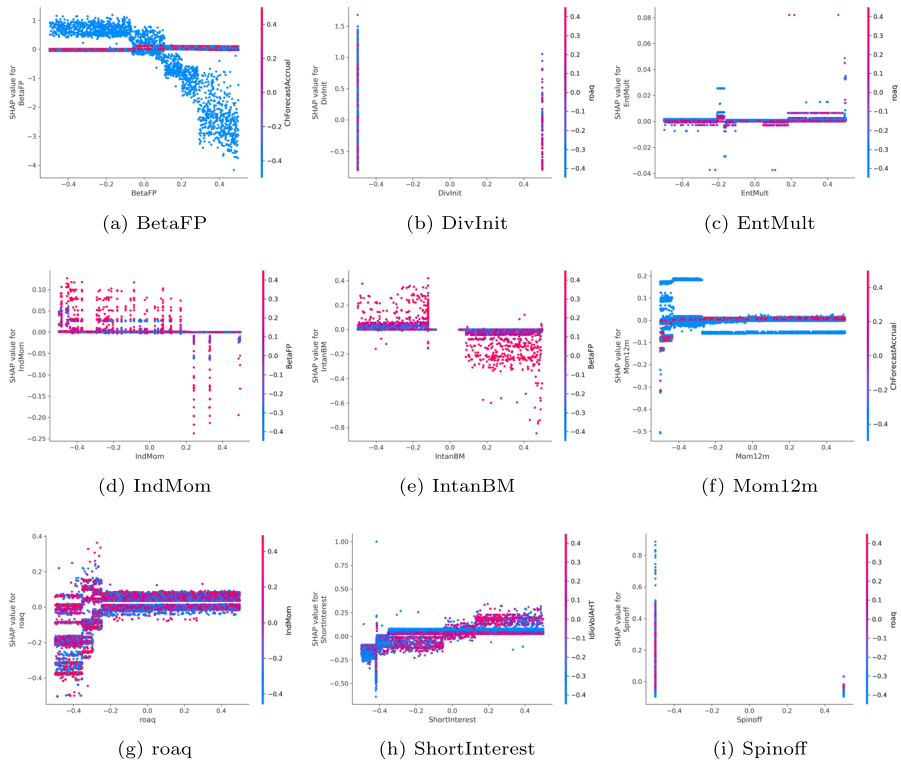


Fig. 18 This figure depicts the dependence plots of the top nine features for the XGB Model, showing their individual and interaction effects in predicting excess return after controlling size, value, momentum, and liquidity (*w/FF4*)

Table 8 Top ten characteristics for each dependent variable and ML model combination

Model	No control	w/ R_m	w/ $FF3$	w/ $FF4$
XTR	IndIPO_1	roaq	Oscore	IndRetBig
	roaq	Beta	MaxRet	roaq
	IdioRisk	ExchSwitch_1	IndRetBig	IdioVolAHT
	IdioVolAHT	DivOmit_1	IdioVol3F	DivInit_3
	DivOmit_2	IdioRisk	IdioVolAHT	Oscore
	MaxRet	MaxRet	std_turn	std_turn
	CF	DivInit_1	cfp	IdioVolAHT_3
	BidAskSpread	ConvDebt_1	roaq	Mom12m
	IndRetBig	CF	Mom12m	MaxRet
	High52	OScore_1	Beta	CBOperProf
RF	ChForecastAccural	IdioVolAHT	ShareVol	IdioVolAHT
	DivInit_2	roaq	ConsRecomm_1	roaq
	SurpriseRD_2	IdioRisk	roaq	ConsRecomm_3
	Spinoff_2	DivOmit_3	ConsRecomm_2	iomom_supp_3
	ShareVol_2	DivOmit_1	IdioVol3F	MaxRet
	CredRatDG_1	UpRecomm	Beta_3	IdioVolAHT_3
	Spinoff	ChForecastAccural_2	iomom_cust_1	Mom6m_2
	UpRecomm_2	ChForecastAccural_3	ExchSwitch_1	Beta_2
	ShareRepurchase_2	Surprise_RD	Beta	Mom12mOffSeason_1
	DivInit	iomom_cust_1	ConsRecomm_3	CF
XGB	Spinoff_1	ExchSwitch_3	ConsRecomm_1	CredRatDG_2
	DivInit_3	DownRecomm_2	NumEarnIncrease	ChForecastAccural_3
	SurpriseRD_2	ConvDebt_2	roaq	DivInit
	ChForecastAccural	OScore_3	AnalystValue	IdioVolAHT
	ConvDebt_1	ConvDebt_3	IdioVolAHT_3	skew1_2
	Spinoff_2	roaq	DivOmit_1	Spinoff
	ShortInterest_1	IdioVolAHT	IdioVolAHT	IndRetBig_3
	DivInit_2	UpRecomm	iomom_supp_1	ConsRecomm_1
	ShortInterest	DownRecomm_3	IdioVol3F	SmileSlope_1
	ShareVol	DivInit	CredRatDG_2	DivOmit_1
NN4	AnnouncementReturn	AnnouncementReturn	AnnouncementReturn	IndRetBig
	DivOmit_2	BidAskSpread	sd_turn	IndRetBig_1
	DivOmit_1	CBOperProf	MaxRet	Beta_3
	DivOmit_3	DivOmit_3	ResidualMomentum	IndRetBig_2
	Frontier	CosSkewness_1	ResidualMomentum_1	CF_1
	Spinoff_3	RD	NetDebtPrice	BetaLiquidityPS_2
	IndMom_3	MaxRet	ResidualMomentum_3	MomSeason06YrPlus_3
	IndMom	DivOmit_1	EntMult	sd_turn
	RD	roaq	roaq	DebtIssuance_2
	NetEquityFinance	High52_1	NumEarnIncrease	Spinoff

Table 8 (continued)

Model	No control	w/ R_m	w/ $FF3$	w/ $FF4$
NN5	Spinoff_3	roaq	roaq	ExchSwitch_3
	IndMom_3	AnnouncementReturn	AnnouncementReturn	NumEarnIncrease
	SmileSlope	zerotrade_2	Spinoff_3	RIO_turnover
	DivOmit_2	DivOmit_1	NetEquityFinance_2	InvestPPEInv_2
	EntMult	Frontier	VolMkt	skew1_2
	roaq	ExchSwitch_1	ShortInterest	Frontier
	IndMom	AnalystRevision	ResidualMomentum_2	ConsRecomm_1
	Spinoff	CredRatDG_1	NetDebtPrice	realestate
	CBOperProf	CBOperProf	std_turn_2	EquityDuration
	Comp_Eqlss	EarningsSurprise_1	Coskewness_2	DivInit_2

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there are no conflict of interest.

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